3,332

No. 10/5

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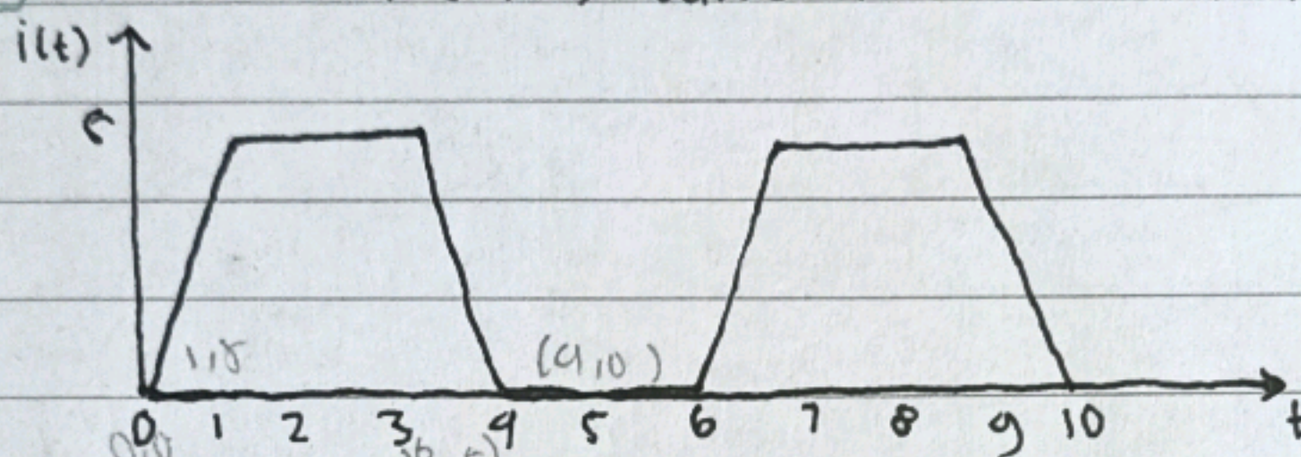
TKM RL 2 AC POWER ANALYSIS

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1A TTE

11.33 Determine the rms valve for the waveform in Fig 11.64



$y - y_1 = m(x - x_1)$

$m = \frac{0-5}{0-1} = 5$

$\Rightarrow I_{rms} = \sqrt{\frac{1}{T}\int_0^T |i(t)|^2 dt}$

$T(\text{periode}) = 6$

$i(t) = 5t \quad 0 < t < 1 \rightarrow i(t) = y_1 = m(t - x_1)$ ⇒ $i(t) = 5t$

$i_2(t) = 5 \quad 1 < t < 3 \rightarrow I_{2(t)} = 5$, $i(t) - y_1 = m(t - x_1)$, $= -5 = 0(t-1)$, $i(t) = 5$

$i_3(t) = -5t + 20 \quad 3 < t < 4 \rightarrow i(t) - t = -5(t-3)$, $i(t) = -5t + 20$

$i_4(t) = 0 \quad 4 < t < 6$

$$I_{rms} = \sqrt{\frac{1}{6}\left[\int_0^1 i_1^2(t)\,dt + \int_1^3 i_2^2\,dt + \int_3^4 i_3^2 + \int_4^6 i_4^2\,dt\right]}$$

$$= \sqrt{\frac{1}{6}\left[\int_0^1 25t^2\,dt + \int_1^3 5^2\,dt + \int_3^4 (25t^2 - 200t + 400)\,dt + \int_4^6 0^2\,dt\right]}$$

$$= \sqrt{\frac{1}{6}\left[\frac{25}{3}|t^3|_0^1 + 25|t|_1^3 + 25\left[\frac{t^3}{3}\Big|_3^4 + 400(t)_3^4 + 0\right]\right]}$$

$$= \sqrt{\frac{1}{6}\left[\frac{25}{3}(1-0) + 25(3-1) + \frac{25}{3}(4^3-3^3) - \frac{200}{2}[4^2-3^2] + 400(4-3)\right]}$$

$$= \sqrt{\frac{1}{6}\left[\frac{25}{3} + (25.2) + \frac{25}{3}(64-27) - 100(16-9) + 400\right]}$$

$$= \sqrt{\frac{1}{6}\left[\frac{25}{3} + 50 + 308,21 - 700 + 400\right]} = \sqrt{\frac{1}{6}\left[66,54\right]}$$

$I_{rms} = \sqrt{11,09} = 3,33017$

atau

$= 3,33\ A$ //